

Question Generation

1 Template

Had **{event}** occurred at time **{t}** ms, what would be the expected values of **{joint_signal}** at $T + \mathbf{\{n\}}$ ms in this counterfactual case?

2 Slot Binding

event $\leftarrow$ +0.5 kg payload
t $\leftarrow$ 4 329
joint_signal $\leftarrow$ joint accel.
n $\leftarrow$ 500

3 Question

Had **an increase of 0.5 kg to the payload** occurred at time **4 329** ms, what would be the expected values of **joint accelerations** at $T + \mathbf{500}$ ms in this counterfactual case?

Answer Generation

4 Answer Rules

A: "Accel. spike above threshold"
rule: $\max(\text{accel}_{[t:t+n]}) > \mu + 2\sigma$ ✓
B: "Torque within nominal range"
rule: $\max(\text{torque}_{[t:t+n]}) < \text{limit} \times$

5 Ground Truth

Answer: {A} ✓
Score: deterministic
(from labels + rules)